

$$\Rightarrow \text{LHS} = \frac{1}{8} \{ \cos 80^\circ - \cos 80^\circ + \cos 60^\circ \} \quad \left[\begin{array}{l} \because \cos (180^\circ - \theta) = -\cos \theta \\ \therefore \cos (180^\circ - 80^\circ) = -\cos 80^\circ \end{array} \right]$$

$$\Rightarrow \text{LHS} = \frac{1}{8} \left\{ \cos 80^\circ - \cos 80^\circ + \frac{1}{2} \right\} = \frac{1}{8} \times \frac{1}{2} = \frac{1}{16} = \text{RHS}$$

EXAMPLE 4 Prove that: $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ = \frac{1}{16}$.

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \sin 30^\circ (\sin 10^\circ \sin 50^\circ) \sin 70^\circ \\ \Rightarrow \text{LHS} &= \frac{1}{2} (\sin 50^\circ \sin 10^\circ) \sin 70^\circ \\ \Rightarrow \text{LHS} &= \frac{1}{2} \times \frac{1}{2} (2 \sin 50^\circ \sin 10^\circ) \sin 70^\circ \\ \Rightarrow \text{LHS} &= \frac{1}{4} \left\{ (2 \sin 50^\circ \sin 10^\circ) \sin 70^\circ \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{4} \left\{ [\cos (50^\circ - 10^\circ) - \cos (50^\circ + 10^\circ)] \sin 70^\circ \right\} \\ &\quad [\because 2 \sin A \sin B = \cos (A - B) - \cos (A + B)] \\ \Rightarrow \text{LHS} &= \frac{1}{4} \left[(\cos 40^\circ - \cos 60^\circ) \sin 70^\circ \right] \\ \Rightarrow \text{LHS} &= \frac{1}{4} \left[\sin 70^\circ \cos 40^\circ - \sin 70^\circ \cos 60^\circ \right] \\ \Rightarrow \text{LHS} &= \frac{1}{4} \left[\sin 70^\circ \cos 40^\circ - \frac{1}{2} \sin 70^\circ \right] \\ \Rightarrow \text{LHS} &= \frac{1}{8} \left[2 \sin 70^\circ \cos 40^\circ - \sin 70^\circ \right] \\ \Rightarrow \text{LHS} &= \frac{1}{8} \left[\sin (70^\circ + 40^\circ) + \sin (70^\circ - 40^\circ) - \sin 70^\circ \right] \\ &\quad [\because 2 \sin A \cos B = \sin (A + B) + \sin (A - B)] \\ \Rightarrow \text{LHS} &= \frac{1}{8} \left\{ \sin 110^\circ + \sin 30^\circ - \sin 70^\circ \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{8} \left\{ \sin (180^\circ - 70^\circ) + \sin 30^\circ - \sin 70^\circ \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{8} \left\{ \sin 70^\circ + \frac{1}{2} - \sin 70^\circ \right\} \quad \left[\because \sin (180 - \theta) = \sin \theta \therefore \sin (180^\circ - 70^\circ) = \sin 70^\circ \right] \\ \Rightarrow \text{LHS} &= \frac{1}{8} \times \frac{1}{2} = \frac{1}{16} = \text{RHS} \end{aligned}$$

ALITER We have,

$$\begin{aligned} \text{LHS} &= \sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ \\ \Rightarrow \text{LHS} &= \sin (90^\circ - 80^\circ) \sin (90^\circ - 60^\circ) \sin (90^\circ - 40^\circ) \sin (90^\circ - 20^\circ) \\ \Rightarrow \text{LHS} &= \cos 80^\circ \cos 60^\circ \cos 40^\circ \cos 20^\circ \\ \Rightarrow \text{LHS} &= \cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16} = \text{RHS} \end{aligned}$$

[See Ex. 3]

EXAMPLE 5 Prove that: $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ = \frac{3}{16}$

SOLUTION We have,

$$\text{LHS} = \sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ$$

$$\Rightarrow \text{LHS} = \sin 60^\circ (\sin 20^\circ \sin 40^\circ) \sin 80^\circ$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3}}{2} \times \frac{1}{2} (2 \sin 20^\circ \sin 40^\circ) \sin 80^\circ$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3}}{4} \left[\cos (40^\circ - 20^\circ) - \cos (40^\circ + 20^\circ) \right] \sin 80^\circ$$

$$[\because 2 \sin A \sin B = \cos (A - B) - \cos (A + B)]$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3}}{4} \left[(\cos 20^\circ - \cos 60^\circ) \sin 80^\circ \right]$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3}}{4} \left[\left(\cos 20^\circ - \frac{1}{2} \right) \sin 80^\circ \right]$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3}}{8} \left[2 \cos 20^\circ \sin 80^\circ - \sin 80^\circ \right]$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3}}{8} \left[\sin (80^\circ + 20^\circ) + \sin (80^\circ - 20^\circ) - \sin 80^\circ \right]$$

$$[\because 2 \sin A \cos B = \sin (A + B) + \sin (A - B)]$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3}}{8} \left\{ \sin 100^\circ + \sin 60^\circ - \sin 80^\circ \right\}$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3}}{8} \left\{ \sin (180^\circ - 80^\circ) + \frac{\sqrt{3}}{2} - \sin 80^\circ \right\}$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3}}{8} \left\{ \sin 80^\circ + \frac{\sqrt{3}}{2} - \sin 80^\circ \right\}$$

$$[\because \sin (180^\circ - 80^\circ) = \sin 80^\circ]$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3}}{8} \times \frac{\sqrt{3}}{2} = \frac{3}{16} = \text{RHS}$$

EXAMPLE 6 Prove that: $4 \cos 12^\circ \cos 48^\circ \cos 72^\circ = \cos 36^\circ$

SOLUTION We have,

$$\text{LHS} = 4 \cos 12^\circ \cos 48^\circ \cos 72^\circ$$

$$\Rightarrow \text{LHS} = 2 (2 \cos 12^\circ \cos 48^\circ) \cos 72^\circ$$

$$\Rightarrow \text{LHS} = 2 (\cos 60^\circ + \cos 36^\circ) \cos 72^\circ$$

$$\Rightarrow \text{LHS} = 2 \cos 60^\circ \cos 72^\circ + 2 \cos 36^\circ \cos 72^\circ$$

$$\Rightarrow \text{LHS} = \cos 72^\circ + \cos 108^\circ + \cos 36^\circ$$

$$\Rightarrow \text{LHS} = \cos 72^\circ + \cos (180^\circ - 72^\circ) + \cos 36^\circ$$

$$\Rightarrow \text{LHS} = \cos 72^\circ - \cos 72^\circ + \cos 36^\circ$$

$$\Rightarrow \text{LHS} = \cos 36^\circ = \text{RHS}$$

EXAMPLE 7 Prove that: $\tan 20^\circ \tan 40^\circ \tan 80^\circ = \tan 60^\circ$

SOLUTION We have,

$$\text{LHS} = \tan 20^\circ \tan 40^\circ \tan 80^\circ = \frac{\sin 20^\circ \sin 40^\circ \sin 80^\circ}{\cos 20^\circ \cos 40^\circ \cos 80^\circ}$$

$$\Rightarrow \text{LHS} = \frac{(2 \sin 20^\circ \sin 40^\circ) \sin 80^\circ}{(2 \cos 20^\circ \cos 40^\circ) \cos 80^\circ} = \frac{(\cos 20^\circ - \cos 60^\circ) \sin 80^\circ}{(\cos 60^\circ + \cos 20^\circ) \cos 80^\circ}$$

$$\Rightarrow \text{LHS} = \frac{\sin 80^\circ \cos 20^\circ - (1/2) \sin 80^\circ}{(1/2) \cos 80^\circ + \cos 80^\circ \cos 20^\circ} = \frac{2 \sin 80^\circ \cos 20^\circ - \sin 80^\circ}{\cos 80^\circ + 2 \cos 80^\circ \cos 20^\circ}$$

$$\Rightarrow \text{LHS} = \frac{\sin 100^\circ + \sin 60^\circ - \sin 80^\circ}{\cos 80^\circ + \cos 100^\circ + \cos 60^\circ} = \frac{\sin (180^\circ - 80^\circ) + \sin 60^\circ - \sin 80^\circ}{\cos 80^\circ + \cos (180^\circ - 80^\circ) + \cos 60^\circ}$$

$$\Rightarrow \text{LHS} = \frac{\sin 80^\circ + \sin 60^\circ - \sin 80^\circ}{\cos 80^\circ - \cos 80^\circ + \cos 60^\circ} = \frac{\sin 60^\circ}{\cos 60^\circ} = \tan 60^\circ = \text{RHS}$$

EXAMPLE 8 Prove that: $\sin A \sin (60^\circ - A) \sin (60^\circ + A) = \frac{1}{4} \sin 3A$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \sin A \sin (60^\circ - A) \sin (60^\circ + A) \\ &= \frac{1}{2} \sin A \left\{ 2 \sin (60^\circ - A) \sin (60^\circ + A) \right\} \\ &= \frac{1}{2} \sin A \left[\cos \left\{ (60^\circ - A) - (60^\circ + A) \right\} - \cos \left\{ (60^\circ - A) + (60^\circ + A) \right\} \right] \\ &= \frac{1}{2} \sin A \left\{ \cos (-2A) - \cos 120^\circ \right\} \\ &= \frac{1}{2} \sin A \left\{ \cos 2A + \frac{1}{2} \right\} \\ &= \frac{1}{2} \sin A \cos 2A + \frac{1}{4} \sin A \\ &= \frac{1}{4} (2 \sin A \cos 2A) + \frac{1}{4} \sin A \\ &= \frac{1}{4} \left\{ \sin (A + 2A) + \sin (A - 2A) \right\} + \frac{1}{4} \sin A \\ &= \frac{1}{4} \left\{ \sin 3A + \sin (-A) \right\} + \frac{1}{4} \sin A \\ &= \frac{1}{4} \sin 3A - \frac{1}{4} \sin A + \frac{1}{4} \sin A = \frac{1}{4} \sin 3A = \text{RHS} \end{aligned}$$

EXAMPLE 9 Prove that: $\cos A \cos (60^\circ - A) \cos (60^\circ + A) = \frac{1}{4} \cos 3A$.

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \cos A \cos (60^\circ - A) \cos (60^\circ + A) \\ &= \frac{1}{2} \cos A \left\{ 2 \cos (60^\circ - A) \cos (60^\circ + A) \right\} \\ &= \frac{1}{2} \cos A \left[\cos \left\{ (60^\circ - A) + (60^\circ + A) \right\} + \cos \left\{ (60^\circ - A) - (60^\circ + A) \right\} \right] \\ &= \frac{1}{2} \cos A \left\{ \cos 120^\circ + \cos (-2A) \right\} \\ &= \frac{1}{2} \cos A \left\{ -\frac{1}{2} + \cos 2A \right\} \\ &= -\frac{1}{4} \cos A + \frac{1}{2} \cos A \cos 2A \\ &= -\frac{1}{4} \cos A + \frac{1}{4} (2 \cos 2A \cos A) \\ &= -\frac{1}{4} \cos A + \frac{1}{4} \left\{ \cos (2A + A) + \cos (2A - A) \right\} \end{aligned}$$

$$\begin{aligned}
 &= -\frac{1}{4} \cos A + \frac{1}{4} (\cos 3A + \cos A) \\
 &= \frac{1}{4} \cos 3A = \text{RHS.}
 \end{aligned}$$

LEVEL-2

EXAMPLE 10 Prove that: $4 \sin \theta \sin \left(\frac{\pi}{3} + \theta \right) \sin \left(\frac{2\pi}{3} + \theta \right) = \sin 3\theta$

SOLUTION We have,

$$\begin{aligned}
 \text{LHS} &= 4 \sin \theta \sin \left(\frac{\pi}{3} + \theta \right) \sin \left(\frac{2\pi}{3} + \theta \right) \\
 &= 2 \sin \theta \left\{ 2 \sin \left(\frac{2\pi}{3} + \theta \right) \sin \left(\frac{\pi}{3} + \theta \right) \right\} \\
 &= 2 \sin \theta \left[\cos \left\{ \left(\frac{2\pi}{3} + \theta \right) - \left(\frac{\pi}{3} + \theta \right) \right\} - \cos \left\{ \left(\frac{2\pi}{3} + \theta \right) + \left(\frac{\pi}{3} + \theta \right) \right\} \right] \\
 &= 2 \sin \theta \left\{ \cos \frac{\pi}{3} - \cos (\pi + 2\theta) \right\} \\
 &= 2 \sin \theta \left\{ \frac{1}{2} + \cos 2\theta \right\} \\
 &= \sin \theta + 2 \sin \theta \cos 2\theta \\
 &= \sin \theta + \{ \sin (\theta + 2\theta) + \sin (\theta - 2\theta) \} \\
 &= \sin \theta + \sin 3\theta + \sin (-\theta) = \sin \theta + \sin 3\theta - \sin \theta = \sin 3\theta = \text{RHS}
 \end{aligned}$$

EXAMPLE 11 Show that: $\tan (60^\circ + \theta) \tan (60^\circ - \theta) = \frac{2 \cos 2\theta + 1}{2 \cos 2\theta - 1}$

SOLUTION We have

LHS

$$\begin{aligned}
 &= \frac{\cos 2\theta - \cos 120^\circ}{\cos 120^\circ + \cos 2\theta} = \frac{\cos 2\theta + \frac{1}{2}}{-\frac{1}{2} + \cos 2\theta} = \frac{2 \cos 2\theta + 1}{2 \cos 2\theta - 1} = \text{RHS}
 \end{aligned}$$

EXAMPLE 12 If $\alpha + \beta = 90^\circ$, find the maximum and minimum values of $\sin \alpha \sin \beta$.

SOLUTION Let $y = \sin \alpha \sin \beta$. Then,

$$y = \frac{1}{2} (2 \sin \alpha \sin \beta)$$

$$\Rightarrow y = \frac{1}{2} \{ \cos (\alpha - \beta) - \cos (\alpha + \beta) \} = \frac{1}{2} \{ \cos (\alpha - \beta) - \cos 90^\circ \} = \frac{1}{2} \cos (\alpha - \beta)$$

We know that

$$-1 \leq \cos (\alpha - \beta) \leq 1$$

$$\Rightarrow \frac{-1}{2} \leq \frac{1}{2} \cos (\alpha - \beta) \leq \frac{1}{2}$$

$$\Rightarrow \frac{-1}{2} \leq y \leq \frac{1}{2}$$

$$\Rightarrow \frac{-1}{2} \leq \sin \alpha \sin \beta \leq \frac{1}{2}$$

Hence, $-\frac{1}{2}$ and $\frac{1}{2}$ are respectively the minimum and maximum values of $\sin \alpha \sin \beta$.

EXERCISE 8.1**LEVEL-1**

1. Express each of the following as the sum or difference of sines and cosines:

(i) $2 \sin 3\theta \cos \theta$

(ii) $2 \cos 3\theta \sin 2\theta$

(iii) $2 \sin 4\theta \sin 3\theta$

(iv) $2 \cos 7\theta \cos 3\theta$

2. Prove that:

(i) $2 \sin \frac{5\pi}{12} \sin \frac{\pi}{12} = \frac{1}{2}$

(ii) $2 \cos \frac{5\pi}{12} \cos \frac{\pi}{12} = \frac{1}{2}$

(iii) $2 \sin \frac{5\pi}{12} \cos \frac{\pi}{12} = \frac{\sqrt{3} + 2}{2}$

3. Show that:

(i) $\sin 50^\circ \cos 85^\circ = \frac{1 - \sqrt{2} \sin 35^\circ}{2\sqrt{2}}$ (ii) $\sin 25^\circ \cos 115^\circ = \frac{1}{2} (\sin 140^\circ - 1)$

4. Prove that: $4 \cos \theta \cos \left(\frac{\pi}{3} + \theta \right) \cos \left(\frac{\pi}{3} - \theta \right) = \cos 3\theta$

5. Prove that :

(i) $\cos 10^\circ \cos 30^\circ \cos 50^\circ \cos 70^\circ = \frac{3}{16}$

(ii) $\cos 40^\circ \cos 80^\circ \cos 160^\circ = -\frac{1}{8}$

(iii) $\sin 20^\circ \sin 40^\circ \sin 80^\circ = \frac{\sqrt{3}}{8}$

(iv) $\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8}$

(v) $\tan 20^\circ \tan 40^\circ \tan 60^\circ \tan 80^\circ = 3$

(vi) $\tan 20^\circ \tan 30^\circ \tan 40^\circ \tan 80^\circ = 1$

(vii) $\sin 10^\circ \sin 50^\circ \sin 60^\circ \sin 70^\circ = \frac{\sqrt{3}}{16}$

(viii) $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ = \frac{3}{16}$

6. Show that :

(i) $\sin A \sin (B - C) + \sin B \sin (C - A) + \sin C \sin (A - B) = 0$

(ii) $\sin (B - C) \cos (A - D) + \sin (C - A) \cos (B - D) + \sin (A - B) \cos (C - D) = 0$

LEVEL-2

7. Prove that : $\tan \theta \tan (60^\circ - \theta) \tan (60^\circ + \theta) = \tan 3\theta$.

8. If $\alpha + \beta = 90^\circ$, show that the maximum value of $\cos \alpha \cos \beta$ is $\frac{1}{2}$.

ANSWERS

1. (i) $\sin 4\theta + \sin 2\theta$ (ii) $\sin 5\theta - \sin \theta$ (iii) $\cos \theta - \cos 7\theta$ (iv) $\cos 10\theta + \cos 4\theta$

HINTS TO SELECTED PROBLEM

2. (i) $\text{LHS} = \cos \left(\frac{5\pi}{12} - \frac{\pi}{12} \right) - \cos \left(\frac{5\pi}{12} + \frac{\pi}{12} \right) = \cos \frac{\pi}{3} - \cos \frac{\pi}{2} = \frac{1}{2} = \text{RHS}$

(ii) $\text{LHS} = \cos \left(\frac{5\pi}{12} + \frac{\pi}{12} \right) + \cos \left(\frac{5\pi}{12} - \frac{\pi}{12} \right) = \cos \frac{\pi}{2} + \cos \frac{\pi}{3} = \frac{1}{2} = \text{RHS}$

(iii) $\text{LHS} = \sin \left(\frac{5\pi}{12} + \frac{\pi}{12} \right) + \sin \left(\frac{5\pi}{12} - \frac{\pi}{12} \right) = \sin \frac{\pi}{2} + \sin \frac{\pi}{3} = 1 + \frac{\sqrt{3}}{2} = \text{RHS}$

8.3 FORMULAE TO TRANSFORM THE SUM OR DIFFERENCE INTO PRODUCT

In the previous section, we have used the following formulae:

$$\begin{aligned}\sin(A+B) + \sin(A-B) &= 2 \sin A \cos B, & \sin(A+B) - \sin(A-B) &= 2 \cos A \sin B \\ \cos(A+B) + \cos(A-B) &= 2 \cos A \cos B \text{ and, } & \cos(A-B) - \cos(A+B) &= 2 \sin A \sin B.\end{aligned}$$

Let $A+B=C$ and $A-B=D$. Then, $A = \frac{C+D}{2}$ and $B = \frac{C-D}{2}$.

Substituting the values of A, B, C and D in the above formulae, we get

$$\sin C + \sin D = 2 \sin \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right) \quad \dots(i)$$

$$\sin C - \sin D = 2 \sin \left(\frac{C-D}{2} \right) \cos \left(\frac{C+D}{2} \right) \quad \dots(ii)$$

$$\cos C + \cos D = 2 \cos \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right) \quad \dots(iii)$$

$$\cos D - \cos C = 2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$$

$$\text{or, } \cos C - \cos D = -2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right) \quad \dots(iv)$$

$$\text{or, } \cos C - \cos D = 2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{D-C}{2} \right)$$

These four formulae are used to convert the sum or difference of two sines or two cosines into the product of sines and cosines.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Express each of the following as a product:

(i) $\sin 4\theta + \sin 2\theta$

(ii) $\sin 6\theta - \sin 2\theta$

(iii) $\cos 4\theta + \cos 8\theta$

(iv) $\cos 6\theta - \cos 8\theta$

SOLUTION (i) $\sin 4\theta + \sin 2\theta$

$$= 2 \sin \left(\frac{4\theta + 2\theta}{2} \right) \cos \left(\frac{4\theta - 2\theta}{2} \right) \quad \left[\because \sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2} \right]$$

$$= 2 \sin 3\theta \cos \theta$$

(ii) $\sin 6\theta - \sin 2\theta$

$$= 2 \sin \left(\frac{6\theta - 2\theta}{2} \right) \cos \left(\frac{6\theta + 2\theta}{2} \right) \quad \left[\because \sin C - \sin D = 2 \sin \left(\frac{C-D}{2} \right) \cos \left(\frac{C+D}{2} \right) \right]$$

$$= 2 \sin 2\theta \cos 4\theta$$

(iii) $\cos 4\theta + \cos 8\theta$

$$= 2 \cos \left(\frac{8\theta + 4\theta}{2} \right) \cos \left(\frac{8\theta - 4\theta}{2} \right) \quad \left[\because \cos C + \cos D = 2 \cos \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right) \right]$$

$$= 2 \cos 6\theta \cos 2\theta$$

(iv) $\cos 6\theta - \cos 8\theta$

$$= 2 \sin \left(\frac{6\theta + 8\theta}{2} \right) \sin \left(\frac{8\theta - 6\theta}{2} \right) \quad \left[\because \cos C - \cos D = 2 \sin \frac{C+D}{2} \sin \frac{D-C}{2} \right]$$

$$= 2 \sin 7\theta \sin \theta$$

EXAMPLE 2 Prove that: $\cos 18^\circ - \sin 18^\circ = \sqrt{2} \sin 27^\circ$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \cos 18^\circ - \sin 18^\circ \\ &= \cos 18^\circ - \cos 72^\circ \quad [\because \sin 18^\circ = \sin (90^\circ - 72^\circ) = \cos 72^\circ] \\ &= 2 \sin \left(\frac{18^\circ + 72^\circ}{2} \right) \sin \left(\frac{72^\circ - 18^\circ}{2} \right) = 2 \sin 45^\circ \sin 27^\circ = \sqrt{2} \sin 27^\circ = \text{RHS} \end{aligned}$$

EXAMPLE 3 Prove that:

$$\begin{aligned} \text{(i)} \quad \frac{\sin 5A - \sin 3A}{\cos 5A + \cos 3A} &= \tan A & \text{(ii)} \quad \frac{\sin A + \sin 3A}{\cos A + \cos 3A} &= \tan 2A & \text{[NCERT]} \\ \text{(iii)} \quad \frac{\sin A + \sin B}{\cos A + \cos B} &= \tan \left(\frac{A+B}{2} \right) & \text{(iv)} \quad \frac{\cos 7A + \cos 5A}{\sin 7A - \sin 5A} &= \cot A & \text{[NCERT]} \end{aligned}$$

SOLUTION (i) $\text{LHS} = \frac{\sin 5A - \sin 3A}{\cos 5A + \cos 3A}$

$$= \frac{2 \sin \left(\frac{5A - 3A}{2} \right) \cos \left(\frac{5A + 3A}{2} \right)}{2 \cos \left(\frac{5A + 3A}{2} \right) \cos \left(\frac{5A - 3A}{2} \right)} = \frac{2 \sin A \cos 4A}{2 \cos 4A \cos A} = \tan A = \text{RHS}$$

(ii) $\text{LHS} = \frac{\sin 3A + \sin A}{\cos 3A + \cos A}$

$$= \frac{2 \sin \left(\frac{3A + A}{2} \right) \cos \left(\frac{3A - A}{2} \right)}{2 \cos \left(\frac{3A + A}{2} \right) \cos \left(\frac{3A - A}{2} \right)} = \frac{\sin 2A \cos A}{\cos 2A \cos A} = \tan 2A = \text{RHS}$$

(iii) $\text{LHS} = \frac{\sin A + \sin B}{\cos A + \cos B} = \frac{2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)}{2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)} = \tan \left(\frac{A+B}{2} \right) = \text{RHS}$

(iv) $\text{LHS} = \frac{\cos 7A + \cos 5A}{\sin 7A - \sin 5A} = \frac{2 \cos \left(\frac{7A + 5A}{2} \right) \cos \left(\frac{7A - 5A}{2} \right)}{2 \sin \left(\frac{7A - 5A}{2} \right) \cos \left(\frac{7A + 5A}{2} \right)} = \frac{2 \cos 6A \cos A}{2 \sin A \cos 6A} = \cot A = \text{RHS}$

EXAMPLE 4 Prove that:

$$\text{(i)} \quad \frac{\cos 9x - \cos 5x}{\sin 17x - \sin 3x} = -\frac{\sin 2x}{\cos 10x} \quad \text{[NCERT]} \quad \text{(ii)} \quad \frac{\sin 5x + \sin 3x}{\cos 5x + \cos 3x} = \tan 4x \quad \text{[NCERT]}$$

$$\text{(iii)} \quad (\sin 3x + \sin x) \sin x + (\cos 3x - \cos x) \cos x = 0 \quad \text{[NCERT]}$$

SOLUTION (i) $\text{LHS} = \frac{\cos 9x - \cos 5x}{\sin 17x - \sin 3x}$

$$= \frac{-2 \sin \left(\frac{9x + 5x}{2} \right) \sin \left(\frac{9x - 5x}{2} \right)}{2 \sin \left(\frac{17x - 3x}{2} \right) \cos \left(\frac{17x + 3x}{2} \right)} = \frac{-2 \sin 7x \sin 2x}{2 \sin 7x \cos 10x} = \frac{-\sin 2x}{\cos 10x} = \text{RHS}$$

(ii) $\text{LHS} = \frac{\sin 5x + \sin 3x}{\cos 5x + \cos 3x}$

$$\frac{2 \sin \left(\frac{5x+3x}{2} \right) \cos \left(\frac{5x-3x}{2} \right)}{2 \cos \left(\frac{5x+3x}{2} \right) \cos \left(\frac{5x-3x}{2} \right)} = \frac{2 \sin 4x \cos x}{2 \cos 4x \cos x} = \tan 4x = \text{RHS}$$

$$\begin{aligned} \text{(iii) LHS} &= (\sin 3x + \sin x) \sin x + (\cos 3x - \cos x) \cos x \\ &= \left\{ 2 \sin \left(\frac{3x+x}{2} \right) \cos \left(\frac{3x-x}{2} \right) \right\} \sin x + \left\{ -2 \sin \left(\frac{3x+x}{2} \right) \sin \left(\frac{3x-x}{2} \right) \right\} \cos x \\ &= 2 \sin 2x \cos x \sin x - 2 \sin 2x \sin x \cos x = 0 = \text{RHS} \end{aligned}$$

EXAMPLE 5 Prove that: $\cot 4x (\sin 5x + \sin 3x) = \cot x (\sin 5x - \sin 3x)$

[NCERT]

$$\begin{aligned} \text{SOLUTION LHS} &= \cot 4x (\sin 5x + \sin 3x) \\ &= \cot 4x \times 2 \sin \left(\frac{5x+3x}{2} \right) \cos \left(\frac{5x-3x}{2} \right) \\ &= \frac{\cos 4x}{\sin 4x} \times 2 \sin 4x \cos x = 2 \cos 4x \cos x \quad \dots \text{(i)} \\ \text{RHS} &= \cot x (\sin 5x - \sin 3x) \\ &= \cot x \times 2 \sin \left(\frac{5x-3x}{2} \right) \cos \left(\frac{5x+3x}{2} \right) \\ &= \frac{\cos x}{\sin x} \times 2 \sin x \cos 4x = 2 \cos 4x \cos x \quad \dots \text{(ii)} \end{aligned}$$

From (i) and (ii), we obtain that LHS = RHS.

EXAMPLE 6 Prove that: $\sin x + \sin 3x + \sin 5x + \sin 7x = 4 \cos x \cos 2x \sin 4x$

[NCERT]

$$\begin{aligned} \text{SOLUTION LHS} &= \sin x + \sin 3x + \sin 5x + \sin 7x \\ &= (\sin 7x + \sin x) + (\sin 5x + \sin 3x) \\ &= 2 \sin \left(\frac{7x+x}{2} \right) \cos \left(\frac{7x-x}{2} \right) + 2 \sin \left(\frac{5x+3x}{2} \right) \cos \left(\frac{5x-3x}{2} \right) \\ &= 2 \sin 4x \cos 3x + 2 \sin 4x \cos x \\ &= 2 \sin 4x (\cos 3x + \cos x) \\ &= 2 \sin 4x \times 2 \cos \left(\frac{3x+x}{2} \right) \cos \left(\frac{3x-x}{2} \right) \\ &= 2 \sin 4x \times 2 \cos 2x \cos x = 4 \cos x \cos 2x \sin 4x = \text{RHS} \end{aligned}$$

EXAMPLE 7 Prove that: $1 + \cos 2x + \cos 4x + \cos 6x = 4 \cos x \cos 2x \cos 3x$

$$\begin{aligned} \text{SOLUTION LHS} &= 1 + \cos 2x + \cos 4x + \cos 6x \\ &= (\cos 0x + \cos 2x) + (\cos 4x + \cos 6x) \\ &= 2 \cos x \cos x + 2 \cos 5x \cos x \\ &= 2 \cos x (\cos x + \cos 5x) \\ &= 2 \cos x (2 \cos 3x \cos 2x) = 4 \cos x \cos 2x \cos 3x = \text{RHS} \end{aligned}$$

EXAMPLE 8 Prove that:

$$\text{(i) } (\sin 3A + \sin A) \sin A + (\cos 3A - \cos A) \cos A = 0$$

$$\text{(ii) } \cos 2\theta \cos \frac{\theta}{2} - \cos 3\theta \cos \frac{9\theta}{2} = \sin 5\theta \sin \frac{5\theta}{2}$$

[NCERT]

$$\text{(iii) } \sin \alpha + \sin \left(\alpha + \frac{2\pi}{3} \right) + \sin \left(\alpha + \frac{4\pi}{3} \right) = 0$$

SOLUTION (i) We have,

$$\text{LHS} = (\sin 3A + \sin A) \sin A + (\cos 3A - \cos A) \cos A$$

$$= \left\{ 2 \sin \left(\frac{3A+A}{2} \right) \cos \left(\frac{3A-A}{2} \right) \right\} \sin A + \left\{ -2 \sin \left(\frac{3A+A}{2} \right) \sin \left(\frac{3A-A}{2} \right) \right\} \cos A$$

$$= 2 \sin 2A \cos A \sin A - 2 \sin 2A \sin A \cos A = 0 = \text{RHS}$$

$$(ii) \quad \text{LHS} = \cos 2\theta \cos \frac{\theta}{2} - \cos 3\theta \cos \frac{9\theta}{2}$$

$$\Rightarrow \text{LHS} = \frac{1}{2} \left\{ 2 \cos 2\theta \cos \frac{\theta}{2} - 2 \cos 3\theta \cos \frac{9\theta}{2} \right\}$$

$$\Rightarrow \text{LHS} = \frac{1}{2} \left[\cos \left\{ \left(2\theta + \frac{\theta}{2} \right) + \cos \left(2\theta - \frac{\theta}{2} \right) \right\} - \left\{ \cos \left(3\theta + \frac{9\theta}{2} \right) + \cos \left(\frac{9\theta}{2} - 3\theta \right) \right\} \right]$$

[Using: $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$]

$$\Rightarrow \text{LHS} = \frac{1}{2} \left\{ \cos \frac{5\theta}{2} + \cos \frac{3\theta}{2} - \cos \frac{15\theta}{2} - \cos \frac{3\theta}{2} \right\}$$

$$\Rightarrow \text{LHS} = \frac{1}{2} \left\{ \cos \frac{5\theta}{2} - \cos \frac{15\theta}{2} \right\}$$

$$\Rightarrow \text{LHS} = \frac{1}{2} \left\{ 2 \sin \left(\frac{\frac{5\theta}{2} + \frac{15\theta}{2}}{2} \right) \sin \left(\frac{\frac{15\theta}{2} - \frac{5\theta}{2}}{2} \right) \right\} \quad \left[\because \cos C - \cos D = 2 \sin \frac{C+D}{2} \sin \frac{D-C}{2} \right]$$

$$\Rightarrow \text{LHS} = \sin 5\theta \sin \frac{5\theta}{2} = \text{RHS}$$

$$(iii) \quad \text{LHS} = \sin \alpha + \sin \left(\alpha + \frac{2\pi}{3} \right) + \sin \left(\alpha + \frac{4\pi}{3} \right)$$

$$\Rightarrow \text{LHS} = \sin \alpha + \left[\sin \left(\alpha + \frac{2\pi}{3} \right) + \sin \left(\alpha + \frac{4\pi}{3} \right) \right]$$

$$\Rightarrow \text{LHS} = \sin \alpha + \left[2 \sin \left(\frac{\alpha + \frac{2\pi}{3} + \alpha + \frac{4\pi}{3}}{2} \right) \cos \left(\frac{\alpha + \frac{4\pi}{3} - \alpha - \frac{2\pi}{3}}{2} \right) \right]$$

$$\Rightarrow \text{LHS} = \sin \alpha + 2 \sin(\alpha + \pi) \cos \frac{\pi}{3} = \sin \alpha + 2(-\sin \alpha) \left(\frac{1}{2} \right) = \sin \alpha - \sin \alpha = 0 = \text{RHS}$$

EXAMPLE 9 Prove that:

$$(i) \quad (\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2 = 4 \cos^2 \left(\frac{\alpha - \beta}{2} \right) \quad [\text{NCERT}]$$

$$(ii) \quad (\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2 = 4 \sin^2 \left(\frac{\alpha - \beta}{2} \right) \quad [\text{NCERT}]$$

$$(iii) \quad \cos \alpha + \cos \beta + \cos \gamma + \cos(\alpha + \beta + \gamma) = 4 \cos \frac{\alpha + \beta}{2} \cos \frac{\beta + \gamma}{2} \cos \frac{\gamma + \alpha}{2}$$

SOLUTION (i) $\text{LHS} = (\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2$

$$\Rightarrow \text{LHS} = \left\{ 2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right) \right\}^2 + \left\{ 2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right) \right\}^2$$

$$\Rightarrow \text{LHS} = 4 \cos^2 \left(\frac{\alpha + \beta}{2} \right) \cos^2 \left(\frac{\alpha - \beta}{2} \right) + 4 \sin^2 \left(\frac{\alpha + \beta}{2} \right) \cos^2 \left(\frac{\alpha - \beta}{2} \right)$$

$$\Rightarrow \text{LHS} = 4 \cos^2 \left(\frac{\alpha - \beta}{2} \right) \left\{ \cos^2 \left(\frac{\alpha + \beta}{2} \right) + \sin^2 \left(\frac{\alpha + \beta}{2} \right) \right\} = 4 \cos^2 \left(\frac{\alpha - \beta}{2} \right) = \text{RHS}$$

(ii)

$$\text{LHS} = (\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2$$

$$\Rightarrow \text{LHS} = \left\{ -2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right) \right\}^2 + \left\{ 2 \sin \left(\frac{\alpha - \beta}{2} \right) \cos \left(\frac{\alpha + \beta}{2} \right) \right\}^2$$

$$\Rightarrow \text{LHS} = 4 \sin^2 \left(\frac{\alpha + \beta}{2} \right) \sin^2 \left(\frac{\alpha - \beta}{2} \right) + 4 \sin^2 \left(\frac{\alpha - \beta}{2} \right) \cos^2 \left(\frac{\alpha + \beta}{2} \right)$$

$$\Rightarrow \text{LHS} = 4 \sin^2 \left(\frac{\alpha - \beta}{2} \right) \left\{ \sin^2 \left(\frac{\alpha + \beta}{2} \right) + \cos^2 \left(\frac{\alpha + \beta}{2} \right) \right\}$$

$$\Rightarrow \text{LHS} = 4 \sin^2 \left(\frac{\alpha - \beta}{2} \right) = \text{RHS} \quad \left[\because \sin^2 \left(\frac{\alpha + \beta}{2} \right) + \cos^2 \left(\frac{\alpha + \beta}{2} \right) = 1 \right]$$

(iii)

$$\text{LHS} = \cos \alpha + \cos \beta + \cos \gamma + \cos (\alpha + \beta + \gamma)$$

$$\Rightarrow \text{LHS} = (\cos \alpha + \cos \beta) + [\cos \gamma + \cos (\alpha + \beta + \gamma)]$$

$$\Rightarrow \text{LHS} = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right) + 2 \cos \left(\frac{\alpha + \beta + \gamma + \gamma}{2} \right) \cos \left(\frac{\alpha + \beta + \gamma - \gamma}{2} \right)$$

$$\Rightarrow \text{LHS} = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right) + 2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha + \beta + 2\gamma}{2} \right)$$

$$\Rightarrow \text{LHS} = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \left\{ \cos \left(\frac{\alpha - \beta}{2} \right) + \cos \left(\frac{\alpha + \beta + 2\gamma}{2} \right) \right\}$$

$$\Rightarrow \text{LHS} = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \left\{ 2 \cos \left(\frac{\frac{\alpha - \beta}{2} + \frac{\alpha + \beta + 2\gamma}{2}}{2} \right) \cos \left(\frac{\frac{\alpha + \beta + 2\gamma}{2} - \frac{\alpha - \beta}{2}}{2} \right) \right\}$$

$$\Rightarrow \text{LHS} = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \left\{ 2 \cos \left(\frac{\alpha + \gamma}{2} \right) \cos \left(\frac{\beta + \gamma}{2} \right) \right\}$$

$$\Rightarrow \text{LHS} = 4 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\beta + \gamma}{2} \right) \cos \left(\frac{\gamma + \alpha}{2} \right) = \text{RHS}$$

EXAMPLE 10 Prove that: $\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x} = \cot 3x$

[NCERT]

SOLUTION

$$\text{LHS} = \frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x}$$

$$\Rightarrow \text{LHS} = \frac{(\cos 4x + \cos 2x) + \cos 3x}{(\sin 4x + \sin 2x) + \sin 3x}$$

$$\Rightarrow \text{LHS} = \frac{2 \cos \left(\frac{4x + 2x}{2} \right) \cos \left(\frac{4x - 2x}{2} \right) + \cos 3x}{2 \sin \left(\frac{4x + 2x}{2} \right) \cos \left(\frac{4x - 2x}{2} \right) + \sin 3x} = \frac{2 \cos 3x \cos x + \cos 3x}{2 \sin 3x \cos x + \sin 3x}$$

$$\Rightarrow \text{LHS} = \frac{\cos 3x (2 \cos x + 1)}{\sin 3x (2 \cos x + 1)} = \frac{\cos 3x}{\sin 3x} = \cot 3x = \text{RHS}$$

EXAMPLE 11 Prove that: $\frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = \tan 4A$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \frac{(\sin 7A + \sin A) + (\sin 5A + \sin 3A)}{(\cos 7A + \cos A) + (\cos 5A + \cos 3A)} \\ &= \frac{2 \sin \left(\frac{7A + A}{2} \right) \cos \left(\frac{7A - A}{2} \right) + 2 \sin \left(\frac{5A + 3A}{2} \right) \cos \left(\frac{5A - 3A}{2} \right)}{2 \cos \left(\frac{7A + A}{2} \right) \cos \left(\frac{7A - A}{2} \right) + 2 \cos \left(\frac{5A + 3A}{2} \right) \cos \left(\frac{5A - 3A}{2} \right)} \\ &= \frac{\sin 4A \cos 3A + \sin 4A \cos A}{\cos 4A \cos 3A + \cos 4A \cos A} = \frac{\sin 4A (\cos 3A + \cos A)}{\cos 4A (\cos 3A + \cos A)} = \tan 4A = \text{RHS} \end{aligned}$$

EXAMPLE 12 Prove that: $\frac{\cos 8A \cos 5A - \cos 12A \cos 9A}{\sin 8A \cos 5A + \cos 12A \sin 9A} = \tan 4A$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \frac{2 \cos 8A \cos 5A - 2 \cos 12A \cos 9A}{2 \sin 8A \cos 5A + 2 \cos 12A \sin 9A} \\ \Rightarrow \text{LHS} &= \frac{\{\cos (8A + 5A) + \cos (8A - 5A)\} - \{\cos (12A + 9A) + \cos (12A - 9A)\}}{\{\sin (8A + 5A) + \sin (8A - 5A)\} + \{\sin (9A + 12A) + \sin (9A - 12A)\}} \\ \Rightarrow \text{LHS} &= \frac{\{\cos 13A + \cos 3A\} - \{\cos 21A + \cos 3A\}}{\{\sin 13A + \sin 3A\} + \{\sin 21A + \sin (-3A)\}} \\ \Rightarrow \text{LHS} &= \frac{(\cos 13A + \cos 3A) - (\cos 21A + \cos 3A)}{(\sin 13A + \sin 3A) + (\sin 21A - \sin 3A)} = \frac{\cos 13A - \cos 21A}{\sin 13A + \sin 21A} \\ \Rightarrow \text{LHS} &= \frac{2 \sin \left(\frac{13A + 21A}{2} \right) \sin \left(\frac{21A - 13A}{2} \right)}{2 \sin \left(\frac{3A + 21A}{2} \right) \cos \left(\frac{21A - 13A}{2} \right)} = \frac{\sin 17A \sin 4A}{\sin 17A \cos 4A} = \tan 4A = \text{RHS} \end{aligned}$$

EXAMPLE 13 Prove that: $\frac{\cos 2A \cos 3A - \cos 2A \cos 7A + \cos A \cos 10A}{\sin 4A \sin 3A - \sin 2A \sin 5A + \sin 4A \sin 7A} = \cot 6A \cot 5A$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \frac{2 \cos 3A \cos 2A - 2 \cos 7A \cos 2A + 2 \cos 10A \cos A}{2 \sin 4A \sin 3A - 2 \sin 5A \sin 2A + 2 \sin 7A \sin 4A} \\ &= \frac{(\cos 5A + \cos A) - (\cos 9A + \cos 5A) + (\cos 11A + \cos 9A)}{(\cos A - \cos 7A) - (\cos 3A - \cos 7A) + (\cos 3A - \cos 11A)} \\ &= \frac{\cos A + \cos 11A}{\cos A - \cos 11A} = \frac{2 \cos \left(\frac{11A + A}{2} \right) \cos \left(\frac{11A - A}{2} \right)}{2 \sin \left(\frac{A + 11A}{2} \right) \sin \left(\frac{11A - A}{2} \right)} \\ &= \frac{\cos 6A \cos 5A}{\sin 6A \sin 5A} = \cot 6A \cot 5A = \text{RHS} \end{aligned}$$

EXAMPLE 14 Prove that: $\frac{\sin (A - C) + 2 \sin A + \sin (A + C)}{\sin (B - C) + 2 \sin B + \sin (B + C)} = \frac{\sin A}{\sin B}$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \frac{\sin(A-C) + \sin(A+C) + 2\sin A}{\sin(B-C) + \sin(B+C) + 2\sin B} \\ &= \frac{2\sin\left(\frac{A-C+A+C}{2}\right)\cos\left(\frac{A+C-A+C}{2}\right) + 2\sin A}{2\sin\left(\frac{B+C+B-C}{2}\right)\cos\left(\frac{B+C-B+C}{2}\right) + 2\sin B} \\ \Rightarrow \text{LHS} &= \frac{2\sin A \cos C + 2\sin A}{2\sin B \cos C + 2\sin B} = \frac{2\sin A (\cos C + 1)}{2\sin B (\cos C + 1)} = \frac{\sin A}{\sin B} = \text{RHS} \end{aligned}$$

LEVEL-2

EXAMPLE 15 If $\sin \theta = n \sin(\theta + 2\alpha)$, prove that $\tan(\theta + \alpha) = \frac{1+n}{1-n} \tan \alpha$.

SOLUTION We have,

$$\begin{aligned} \sin \theta &= n \sin(\theta + 2\alpha) \\ \Rightarrow \frac{\sin(\theta + 2\alpha)}{\sin \theta} &= \frac{1}{n} \\ \Rightarrow \frac{\sin(\theta + 2\alpha) + \sin \theta}{\sin(\theta + 2\alpha) - \sin \theta} &= \frac{1+n}{1-n} && [\text{Applying componendo-dividendo}] \\ \Rightarrow \frac{2\sin(\theta + \alpha)\cos \alpha}{2\sin \alpha \cos(\theta + \alpha)} &= \frac{1+n}{1-n} \\ \Rightarrow \tan(\theta + \alpha) &= \frac{1+n}{1-n} \tan \alpha \end{aligned}$$

EXAMPLE 16 Prove that:

$$\left(\frac{\cos A + \cos B}{\sin A - \sin B}\right)^n + \left(\frac{\sin A + \sin B}{\cos A - \cos B}\right)^n = \begin{cases} 2 \cot^n\left(\frac{A-B}{2}\right) & , \text{ if } n \text{ is even} \\ 0 & , \text{ if } n \text{ is odd} \end{cases}$$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \left(\frac{2\cos\frac{A+B}{2}\cos\frac{A-B}{2}}{2\sin\frac{A-B}{2}\cos\frac{A+B}{2}}\right)^n + \left(\frac{2\sin\frac{A+B}{2}\cos\frac{A-B}{2}}{-2\sin\frac{A+B}{2}\sin\frac{A-B}{2}}\right)^n \\ \Rightarrow \text{LHS} &= \left\{\cot\left(\frac{A-B}{2}\right)\right\}^n + \left\{-\cot\left(\frac{A-B}{2}\right)\right\}^n \\ \Rightarrow \text{LHS} &= \cot^n\left(\frac{A-B}{2}\right) + (-1)^n \cot^n\left(\frac{A-B}{2}\right) \\ \Rightarrow \text{LHS} &= \cot^n\left(\frac{A-B}{2}\right) \left\{1 + (-1)^n\right\} = \begin{cases} 2 \cot^n\left(\frac{A-B}{2}\right) & , \text{ if } n \text{ is even} \\ 0 & , \text{ if } n \text{ is odd} \end{cases} \end{aligned}$$

EXAMPLE 17 If three angles A, B and C are in A.P., prove that: $\cot B = \frac{\sin A - \sin C}{\cos C - \cos A}$.

SOLUTION We have,

$$\text{RHS} = \frac{2 \sin \frac{A-C}{2} \cos \frac{A+C}{2}}{2 \sin \frac{A+C}{2} \sin \frac{A-C}{2}} = \cot \left(\frac{A+C}{2} \right) = \cot B = \text{LHS} \quad \left[\begin{array}{l} \because A, B, C \text{ are in A.P.} \\ \therefore 2B = A + C \end{array} \right]$$

EXAMPLE 18 If $\sin \theta + \sin \phi = \sqrt{3} (\cos \phi - \cos \theta)$, prove that $\sin 3\theta + \sin 3\phi = 0$

SOLUTION We have,

[NCERT EXEMPLAR]

$$\begin{aligned} \sin \theta + \sin \phi &= \sqrt{3} (\cos \phi - \cos \theta) \\ \Rightarrow 2 \sin \frac{\theta + \phi}{2} \cos \frac{\theta - \phi}{2} &= 2\sqrt{3} \sin \frac{\theta + \phi}{2} \sin \frac{\theta - \phi}{2} \\ \Rightarrow \left\{ \cos \frac{\theta - \phi}{2} - \sqrt{3} \sin \frac{\theta - \phi}{2} \right\} \sin \left(\frac{\theta + \phi}{2} \right) &= 0 \\ \Rightarrow \sin \left(\frac{\theta + \phi}{2} \right) = 0 \text{ or, } \cos \left(\frac{\theta - \phi}{2} \right) - \sqrt{3} \sin \left(\frac{\theta - \phi}{2} \right) &= 0 \\ \Rightarrow \sin \left(\frac{\theta + \phi}{2} \right) = 0 \text{ or, } \tan \left(\frac{\theta - \phi}{2} \right) &= \frac{1}{\sqrt{3}} = \tan \frac{\pi}{6} \\ \Rightarrow \frac{\theta + \phi}{2} = 0 \quad \text{or,} \quad \frac{\theta - \phi}{2} = \frac{\pi}{6} \\ \Rightarrow \theta = -\phi \quad \text{or,} \quad \theta - \phi = \frac{\pi}{3} \end{aligned}$$

CASE I When $\theta = -\phi$: In this case, we have

$$\sin 3\theta + \sin 3\phi = \sin 3(-\phi) + \sin 3\phi = -\sin 3\phi + \sin 3\phi = 0$$

CASE II When $\theta - \phi = \frac{\pi}{3}$: In this case, we have

$$\theta - \phi = \frac{\pi}{3} \Rightarrow 3\theta - 3\phi = \pi \Rightarrow 3\theta = \pi + 3\phi$$

$$\therefore \sin 3\theta + \sin 3\phi = \sin (\pi + 3\phi) + \sin 3\phi = -\sin 3\phi + \sin 3\phi = 0$$

EXAMPLE 19 If $\frac{\sin (\theta + \alpha)}{\cos (\theta - \alpha)} = \frac{1 - m}{1 + m}$, prove that $\tan \left(\frac{\pi}{4} - \theta \right) \tan \left(\frac{\pi}{4} - \alpha \right) = m$

SOLUTION We have,

$$\begin{aligned} \frac{\sin (\theta + \alpha)}{\cos (\theta - \alpha)} &= \frac{1 - m}{1 + m} \\ \Rightarrow \frac{\sin (\theta + \alpha) + \cos (\theta - \alpha)}{\sin (\theta + \alpha) - \cos (\theta - \alpha)} &= \frac{2}{-2m} \quad \text{[Using componendo-dividendo]} \\ \Rightarrow \frac{\sin (\theta + \alpha) + \sin \left\{ \frac{\pi}{2} - (\theta - \alpha) \right\}}{\sin (\theta + \alpha) - \sin \left\{ \frac{\pi}{2} - (\theta - \alpha) \right\}} &= -\frac{1}{m} \\ \Rightarrow \frac{2 \sin \left(\frac{\theta + \alpha + \frac{\pi}{2} - \theta + \alpha}{2} \right) \cos \left(\frac{\theta + \alpha - \frac{\pi}{2} + \theta - \alpha}{2} \right)}{2 \sin \left(\frac{\theta + \alpha - \frac{\pi}{2} + \theta - \alpha}{2} \right) \cos \left(\frac{\theta + \alpha + \frac{\pi}{2} - \theta + \alpha}{2} \right)} &= -\frac{1}{m} \end{aligned}$$

$$\Rightarrow \frac{\sin\left(\frac{\pi}{4} + \alpha\right) \cos\left(-\frac{\pi}{4} + \theta\right)}{\sin\left(-\frac{\pi}{4} + \theta\right) \cos\left(\frac{\pi}{4} + \alpha\right)} = -\frac{1}{m}$$

$$\Rightarrow \frac{\sin\left(\frac{\pi}{4} + \alpha\right) \cos\left(\frac{\pi}{4} - \theta\right)}{\cos\left(\frac{\pi}{4} + \alpha\right) \sin\left(\frac{\pi}{4} - \theta\right)} = \frac{1}{m}$$

$$\Rightarrow \tan\left(\frac{\pi}{4} + \alpha\right) \cot\left(\frac{\pi}{4} - \theta\right) = \frac{1}{m}$$

$$\Rightarrow m = \cot\left(\frac{\pi}{4} + \alpha\right) \tan\left(\frac{\pi}{4} - \theta\right)$$

$$\Rightarrow m = \tan\left\{\frac{\pi}{2} - \left(\frac{\pi}{4} + \alpha\right)\right\} \tan\left(\frac{\pi}{4} - \theta\right)$$

$$\Rightarrow m = \tan\left(\frac{\pi}{4} - \alpha\right) \tan\left(\frac{\pi}{4} - \theta\right)$$

EXAMPLE 20 If $a \sin \theta = b \sin\left(\theta + \frac{2\pi}{3}\right) = c \sin\left(\theta + \frac{4\pi}{3}\right)$, prove that $ab + bc + ca = 0$.

SOLUTION We have,

$$a \sin \theta = b \sin\left(\theta + \frac{2\pi}{3}\right) = c \sin\left(\theta + \frac{4\pi}{3}\right) = \lambda \text{ (say)}$$

$$\Rightarrow \frac{\lambda}{a} = \sin \theta, \frac{\lambda}{b} = \sin\left(\theta + \frac{2\pi}{3}\right) \text{ and } \frac{\lambda}{c} = \sin\left(\theta + \frac{4\pi}{3}\right)$$

$$\Rightarrow \frac{\lambda}{a} + \frac{\lambda}{b} + \frac{\lambda}{c} = \sin \theta + \sin\left(\theta + \frac{2\pi}{3}\right) + \sin\left(\theta + \frac{4\pi}{3}\right)$$

$$\Rightarrow \frac{\lambda}{a} + \frac{\lambda}{b} + \frac{\lambda}{c} = \left\{\sin\left(\theta + \frac{4\pi}{3}\right) + \sin \theta\right\} + \sin\left(\theta + \frac{2\pi}{3}\right)$$

$$\Rightarrow \frac{\lambda}{a} + \frac{\lambda}{b} + \frac{\lambda}{c} = 2 \sin\left(\theta + \frac{2\pi}{3}\right) \cos \frac{2\pi}{3} + \sin\left(\theta + \frac{2\pi}{3}\right)$$

$$\Rightarrow \frac{\lambda}{a} + \frac{\lambda}{b} + \frac{\lambda}{c} = -\sin\left(\theta + \frac{2\pi}{3}\right) + \sin\left(\theta + \frac{2\pi}{3}\right) = 0$$

$$\Rightarrow \lambda \left(\frac{bc + ca + ab}{abc}\right) = 0 \Rightarrow ab + bc + ca = 0$$

EXAMPLE 21 If $\sin(y + z - x)$, $\sin(z + x - y)$, $\sin(x + y - z)$ are in A.P., prove that $\tan x$, $\tan y$, $\tan z$ are also in A.P.

SOLUTION It is given that $\sin(y + z - x)$, $\sin(z + x - y)$ and $\sin(x + y - z)$ are in A.P.

$$\therefore \sin(z + x - y) - \sin(y + z - x) = \sin(x + y - z) - \sin(z + x - y)$$

$$\Rightarrow 2 \sin(x - y) \cos z = 2 \sin(y - z) \cos x$$

$$\Rightarrow \sin(x - y) \cos z = \sin(y - z) \cos x$$

$$\Rightarrow \sin x \cos y \cos z - \cos x \sin y \cos z = \sin y \cos z \cos x - \cos y \sin z \cos x$$

$$\Rightarrow 2 \sin y \cos x \cos z = \sin x \cos y \cos z + \cos x \cos y \sin z$$

$$\Rightarrow 2 \tan y = \tan x + \tan z$$

$$\Rightarrow \tan x, \tan y, \tan z \text{ are in A.P.}$$

[Dividing throughout by $\cos x \cos y \cos z$]

EXAMPLE 22 If $\frac{\tan(\theta + \alpha)}{a} = \frac{\tan(\theta + \beta)}{b} = \frac{\tan(\theta + \gamma)}{c}$, prove that

$$\frac{a+b}{a-b} \sin^2(\alpha - \beta) + \frac{b+c}{b-c} \sin^2(\beta - \gamma) + \frac{c+a}{c-a} \sin^2(\gamma - \alpha) = 0$$

SOLUTION We have,

$$\frac{\tan(\theta + \alpha)}{a} = \frac{\tan(\theta + \beta)}{b}$$

$$\Rightarrow \frac{a}{b} = \frac{\tan(\theta + \alpha)}{\tan(\theta + \beta)}$$

$$\Rightarrow \frac{a+b}{a-b} = \frac{\tan(\theta + \alpha) + \tan(\theta + \beta)}{\tan(\theta + \alpha) - \tan(\theta + \beta)}$$

[Applying Componendo-dividendo]

$$\Rightarrow \frac{a+b}{a-b} = \frac{\sin(2\theta + \alpha + \beta)}{\sin(\alpha - \beta)} \quad \left[\because \frac{\tan A + \tan B}{\tan A - \tan B} = \frac{\sin(A+B)}{\sin(A-B)} \right]$$

$$\Rightarrow \frac{a+b}{a-b} \sin^2(\alpha - \beta) = \sin(2\theta + \alpha + \beta) \sin(\alpha - \beta)$$

$$\Rightarrow \frac{a+b}{a-b} \sin^2(\alpha - \beta) = \frac{1}{2} \left\{ 2 \sin(2\theta + \alpha + \beta) \sin(\alpha - \beta) \right\}$$

$$\Rightarrow \frac{a+b}{a-b} \sin^2(\alpha - \beta) = \frac{1}{2} \left\{ \cos(2\theta + 2\beta) - \cos(2\theta + 2\alpha) \right\}$$

Similarly, we obtain

$$\frac{b+c}{b-c} \sin^2(\beta - \gamma) = \frac{1}{2} \left\{ \cos(2\theta + 2\gamma) - \cos(2\theta + 2\beta) \right\}$$

$$\text{and, } \frac{c+a}{c-a} \sin^2(\gamma - \alpha) = \frac{1}{2} \left\{ \cos(2\theta + 2\alpha) - \cos(2\theta + 2\gamma) \right\}$$

$$\therefore \frac{a+b}{a-b} \sin^2(\alpha - \beta) + \frac{b+c}{b-c} \sin^2(\beta - \gamma) + \frac{c+a}{c-a} \sin^2(\gamma - \alpha)$$

$$= \frac{1}{2} \left\{ \cos(2\theta + 2\beta) - \cos(2\theta + 2\alpha) + \cos(2\theta + 2\gamma) - \cos(2\theta + 2\beta) + \cos(2\theta + 2\alpha) - \cos(2\theta + 2\gamma) \right\}$$

$$= \frac{1}{2} \times 0 = 0$$

EXAMPLE 23 Prove that: $\frac{\cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10}{\cos 5\theta + 5 \cos 3\theta + 10 \cos \theta} = 2 \cos \theta$

SOLUTION We have,

$$\cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10$$

$$= (\cos 6\theta + \cos 4\theta) + (5 \cos 4\theta + 5 \cos 2\theta) + (10 \cos 2\theta + 10)$$

$$= (\cos 6\theta + \cos 4\theta) + 5(\cos 4\theta + \cos 2\theta) + 10(\cos 2\theta + \cos 0\theta)$$

$$= 2 \cos 5\theta \cos \theta + 5 \times 2 \cos 3\theta \cos \theta + 10 \times 2 \cos \theta \cos \theta$$

$$= 2 \cos \theta (\cos 5\theta + 5 \cos 3\theta + 10 \cos \theta)$$

$$\therefore \text{LHS} = \frac{\cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10}{\cos 5\theta + 5 \cos 3\theta + 10 \cos \theta}$$

$$\Rightarrow \text{LHS} = \frac{2 \cos \theta (\cos 5\theta + 5 \cos 3\theta + 10 \cos \theta)}{\cos 5\theta + 5 \cos 3\theta + 10 \cos \theta} = 2 \cos \theta = \text{RHS}$$

EXERCISE 8.2

LEVEL-1

1. Express each of the following as the product of sines and cosines:

- (i) $\sin 12\theta + \sin 4\theta$ (ii) $\sin 5\theta - \sin \theta$ (iii) $\cos 12\theta + \cos 8\theta$
 (iv) $\cos 12\theta - \cos 4\theta$ (v) $\sin 2\theta + \cos 4\theta$

2. Prove that:

- (i) $\sin 38^\circ + \sin 22^\circ = \sin 82^\circ$ (ii) $\cos 100^\circ + \cos 20^\circ = \cos 40^\circ$
 (iii) $\sin 50^\circ + \sin 10^\circ = \cos 20^\circ$ (iv) $\sin 23^\circ + \sin 37^\circ = \cos 7^\circ$
 (v) $\sin 105^\circ + \cos 105^\circ = \cos 45^\circ$ (vi) $\sin 40^\circ + \sin 20^\circ = \cos 10^\circ$

3. Prove that:

- (i) $\cos 55^\circ + \cos 65^\circ + \cos 175^\circ = 0$ (ii) $\sin 50^\circ - \sin 70^\circ + \sin 10^\circ = 0$
 (iii) $\cos 80^\circ + \cos 40^\circ - \cos 20^\circ = 0$ (iv) $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ = 0$
 (v) $\sin \frac{5\pi}{18} - \cos \frac{4\pi}{9} = \sqrt{3} \sin \frac{\pi}{9}$ (vi) $\cos \frac{\pi}{12} - \sin \frac{\pi}{12} = \frac{1}{\sqrt{2}}$
 (vii) $\sin 80^\circ - \cos 70^\circ = \cos 50^\circ$ (viii) $\sin 51^\circ + \cos 81^\circ = \cos 21^\circ$

4. Prove that:

$$(i) \cos\left(\frac{3\pi}{4} + x\right) - \cos\left(\frac{3\pi}{4} - x\right) = -\sqrt{2} \sin x \quad \text{[NCERT]}$$

$$(ii) \cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right) = \sqrt{2} \cos x \quad \text{[NCERT]}$$

5. Prove that :

$$(i) \sin 65^\circ + \cos 65^\circ = \sqrt{2} \cos 20^\circ \quad (ii) \sin 47^\circ + \cos 77^\circ = \cos 17^\circ$$

6. Prove that :

- (i) $\cos 3A + \cos 5A + \cos 7A + \cos 15A = 4 \cos 4A \cos 5A \cos 6A$
 (ii) $\cos A + \cos 3A + \cos 5A + \cos 7A = 4 \cos A \cos 2A \cos 4A$
 (iii) $\sin A + \sin 2A + \sin 4A + \sin 5A = 4 \cos \frac{A}{2} \cos \frac{3A}{2} \sin 3A$
 (iv) $\sin 3A + \sin 2A - \sin A = 4 \sin A \cos \frac{A}{2} \cos \frac{3A}{2}$
 (v) $\cos 20^\circ \cos 100^\circ + \cos 100^\circ \cos 140^\circ - \cos 140^\circ \cos 200^\circ = -\frac{3}{4}$
 (vi) $\sin \frac{\theta}{2} \sin \frac{7\theta}{2} + \sin \frac{3\theta}{2} \sin \frac{11\theta}{2} = \sin 2\theta \sin 5\theta$
 (vii) $\cos \theta \cos \frac{\theta}{2} - \cos 3\theta \cos \frac{9\theta}{2} = \sin 7\theta \sin 8\theta$

[NCERT EXEMPLAR]

7. Prove that:

- (i) $\frac{\sin A + \sin 3A}{\cos A - \cos 3A} = \cot A$
 (ii) $\frac{\sin 9A - \sin 7A}{\cos 7A - \cos 9A} = \cot 8A$
 (iii) $\frac{\sin A - \sin B}{\cos A + \cos B} = \tan \frac{A-B}{2}$

$$(iv) \frac{\sin A + \sin B}{\sin A - \sin B} = \tan \left(\frac{A+B}{2} \right) \cot \left(\frac{A-B}{2} \right)$$

$$(v) \frac{\cos A + \cos B}{\cos B - \cos A} = \cot \left(\frac{A+B}{2} \right) \cot \left(\frac{A-B}{2} \right)$$

8. Prove that:

$$(i) \frac{\sin A + \sin 3A + \sin 5A}{\cos A + \cos 3A + \cos 5A} = \tan 3A$$

$$(ii) \frac{\cos 3A + 2 \cos 5A + \cos 7A}{\cos A + 2 \cos 3A + \cos 5A} = \frac{\cos 5A}{\cos 3A}$$

$$(iii) \frac{\cos 4A + \cos 3A + \cos 2A}{\sin 4A + \sin 3A + \sin 2A} = \cot 3A$$

[NCERT]

$$(iv) \frac{\sin 3A + \sin 5A + \sin 7A + \sin 9A}{\cos 3A + \cos 5A + \cos 7A + \cos 9A} = \tan 6A$$

$$(v) \frac{\sin 5A - \sin 7A + \sin 8A - \sin 4A}{\cos 4A + \cos 7A - \cos 5A - \cos 8A} = \cot 6A$$

$$(vi) \frac{\sin 5A \cos 2A - \sin 6A \cos A}{\sin A \sin 2A - \cos 2A \cos 3A} = \tan A$$

$$(vii) \frac{\sin 11A \sin A + \sin 7A \sin 3A}{\cos 11A \sin A + \cos 7A \sin 3A} = \tan 8A$$

$$(viii) \frac{\sin 3A \cos 4A - \sin A \cos 2A}{\sin 4A \sin A + \cos 6A \cos A} = \tan 2A$$

$$(ix) \frac{\sin A \sin 2A + \sin 3A \sin 6A}{\sin A \cos 2A + \sin 3A \cos 6A} = \tan 5A$$

$$(x) \frac{\sin A + 2 \sin 3A + \sin 5A}{\sin 3A + 2 \sin 5A + \sin 7A} = \frac{\sin 3A}{\sin 5A}$$

$$(xi) \frac{\sin (\theta + \phi) - 2 \sin \theta + \sin (\theta - \phi)}{\cos (\theta + \phi) - 2 \cos \theta + \cos (\theta - \phi)} = \tan \theta$$

9. Prove that:

$$(i) \sin \alpha + \sin \beta + \sin \gamma - \sin (\alpha + \beta + \gamma) = 4 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\beta + \gamma}{2} \right) \sin \left(\frac{\gamma + \alpha}{2} \right)$$

$$(ii) \cos (A + B + C) + \cos (A - B + C) + \cos (A + B - C) + \cos (-A + B + C)$$

$$= 4 \cos A \cos B \cos C$$

LEVEL-2

10. If $\cos A + \cos B = \frac{1}{2}$ and $\sin A + \sin B = \frac{1}{4}$, prove that: $\tan \left(\frac{A+B}{2} \right) = \frac{1}{2}$

11. If $\operatorname{cosec} A + \sec A = \operatorname{cosec} B + \sec B$, prove that: $\tan A \tan B = \cot \frac{A+B}{2}$

12. If $\sin 2A = \lambda \sin 2B$, prove that: $\frac{\tan(A+B)}{\tan(A-B)} = \frac{\lambda+1}{\lambda-1}$

13. Prove that:

(i) $\frac{\cos(A+B+C) + \cos(-A+B+C) + \cos(A-B+C) + \cos(A+B-C)}{\sin(A+B+C) + \sin(-A+B+C) + \sin(A-B+C) - \sin(A+B-C)} = \cot C$

(ii) $\sin(B-C) \cos(A-D) + \sin(C-A) \cos(B-D) + \sin(A-B) \cos(C-D) = 0$

14. If $\frac{\cos(A-B)}{\cos(A+B)} + \frac{\cos(C+D)}{\cos(C-D)} = 0$, prove that $\tan A \tan B \tan C \tan D = -1$

15. If $\cos(\alpha + \beta) \sin(\gamma + \delta) = \cos(\alpha - \beta) \sin(\gamma - \delta)$, prove that $\cot \alpha \cot \beta \cot \gamma = \cot \delta$

16. If $y \sin \phi = x \sin(2\theta + \phi)$, prove that $(x+y) \cot(\theta + \phi) = (y-x) \cot \theta$

17. If $\cos(A+B) \sin(C-D) = \cos(A-B) \sin(C+D)$, prove that $\tan A \tan B \tan C + \tan D = 0$

18. If $x \cos \theta = y \cos\left(\theta + \frac{2\pi}{3}\right) = z \cos\left(\theta + \frac{4\pi}{3}\right)$, prove that $xy + yz + zx = 0$.

[NCERT EXEMPLAR]

19. If $m \sin \theta = n \sin(\theta + 2\alpha)$, prove that $\tan(\theta + \alpha) \cot \alpha = \frac{m+n}{m-n}$.

[NCERT EXEMPLAR]

ANSWERS

1. (i) $2 \sin 8\theta \cos 4\theta$ (ii) $2 \sin 2\theta \cos 3\theta$ (iii) $2 \cos 10\theta \cos 2\theta$
 (iv) $-2 \sin 8\theta \sin 4\theta$ (v) $2 \cos\left(\frac{\pi}{4} + \theta\right) \cos\left(\frac{\pi}{4} - 3\theta\right)$

HINTS TO NCERT & SELECTED PROBLEMS

4. (i) $\text{LHS} = \cos\left(\frac{3\pi}{4} + x\right) - \cos\left(\frac{3\pi}{4} - x\right)$

$$= -2 \sin \left\{ \frac{\left(\frac{3\pi}{4} + x\right) + \left(\frac{3\pi}{4} - x\right)}{2} \right\} \sin \left\{ \frac{\left(\frac{3\pi}{4} + x\right) - \left(\frac{3\pi}{4} - x\right)}{2} \right\}$$

$$= -2 \sin \frac{3\pi}{4} \sin x = -2 \times \frac{1}{\sqrt{2}} \sin x = -\sqrt{2} \sin x = \text{RHS}$$

(ii) $\text{LHS} = \cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right)$

$$= 2 \cos \left\{ \frac{\left(\frac{\pi}{4} + x\right) + \left(\frac{\pi}{4} - x\right)}{2} \right\} \cos \left\{ \frac{\left(\frac{\pi}{4} + x\right) - \left(\frac{\pi}{4} - x\right)}{2} \right\}$$

$$= 2 \cos \frac{\pi}{4} \cos x = 2 \times \frac{1}{\sqrt{2}} \cos x = \sqrt{2} \cos x = \text{RHS}$$

8. (iii) $\text{LHS} = \frac{\cos 4A + \cos 3A + \cos 2A}{\sin 4A + \sin 3A + \sin 2A}$

$$= \frac{(\cos 4A + \cos 2A) + \cos 3A}{(\sin 4A + \sin 2A) + \sin 3A}$$

$$= \frac{2 \cos 3A \cos A + \cos 3A}{2 \sin 3A \cos A + \sin 3A} = \frac{\cos 3A (2 \cos A + 1)}{\sin 3A (2 \cos A + 1)} = \cot 3A = \text{RHS}$$

VERY SHORT ANSWER QUESTIONS (VSAQs)

Answer each of the following questions in one word or one sentence or as per exact requirement of the question:

1. If $(\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2 = \lambda \cos^2 \left(\frac{\alpha - \beta}{2} \right)$, write the value of λ .
2. Write the value of $\sin \frac{\pi}{12} \sin \frac{5\pi}{12}$.
3. If $\sin A + \sin B = \alpha$ and $\cos A + \cos B = \beta$, then write the value of $\tan \left(\frac{A+B}{2} \right)$.
4. If $\cos A = m \cos B$, then write the value of $\cot \frac{A+B}{2} \cot \frac{A-B}{2}$.
5. Write the value of the expression $\frac{1 - 4 \sin 10^\circ \sin 70^\circ}{2 \sin 10^\circ}$.
6. If $A + B = \frac{\pi}{3}$ and $\cos A + \cos B = 1$, then find the value of $\cos \frac{A-B}{2}$.
7. Write the value of $\sin 12^\circ \sin 48^\circ \sin 54^\circ$.
8. If $\sin 2A = \lambda \sin 2B$, then write the value of $\frac{\lambda + 1}{\lambda - 1}$.
9. Write the value of $\frac{\sin A + \sin 3A}{\cos A + \cos 3A}$.
10. If $\cos(A+B) \sin(C-D) = \cos(A-B) \sin(C+D)$, then write the value $\tan A \tan B \tan C$.

ANSWERS

- | | | | | | |
|------------------|----------------------------------|---------------------------|----------------------|------|-------------------------|
| 1. 4 | 2. $\frac{1}{2}$ | 3. $\frac{\alpha}{\beta}$ | 4. $\frac{1+m}{1-m}$ | 5. 1 | 6. $\frac{1}{\sqrt{3}}$ |
| 7. $\frac{1}{8}$ | 8. $\frac{\tan(A+B)}{\tan(A-B)}$ | 9. $\tan 2A$ | 10. $-\tan D$ | | |

MULTIPLE CHOICE QUESTIONS (MCQs)

Mark the correct alternative in each of the following:

1. $\cos 40^\circ + \cos 80^\circ + \cos 160^\circ + \cos 240^\circ =$
 (a) 0 (b) 1 (c) $1/2$ (d) $-1/2$
2. $\sin 163^\circ \cos 347^\circ + \sin 73^\circ \sin 167^\circ =$
 (a) 0 (b) $1/2$ (c) 1 (d) none of these
3. If $\sin 2\theta + \sin 2\phi = \frac{1}{2}$ and $\cos 2\theta + \cos 2\phi = \frac{3}{2}$, then $\cos^2(\theta - \phi) =$
 (a) $3/8$ (b) $5/8$ (c) $3/4$ (d) $5/4$
4. The value of $\cos 52^\circ + \cos 68^\circ + \cos 172^\circ$ is
 (a) 0 (b) 1 (c) 2 (d) $3/2$
5. The value of $\sin 78^\circ - \sin 66^\circ - \sin 42^\circ + \sin 6^\circ$ is
 (a) $1/2$ (b) $-1/2$ (c) -1 (d) none of these
6. If $\sin \alpha + \sin \beta = a$ and $\cos \alpha - \cos \beta = b$, then $\tan \frac{\alpha - \beta}{2} =$

- (a) $-\frac{a}{b}$ (b) $-\frac{b}{a}$ (c) $\sqrt{a^2 + b^2}$ (d) none of these
7. $\cos 35^\circ + \cos 85^\circ + \cos 155^\circ =$
 (a) 0 (b) $\frac{1}{\sqrt{3}}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\cos 275^\circ$
8. The value of $\sin 50^\circ - \sin 70^\circ + \sin 10^\circ$ is equal to
 (a) 1 (b) 0 (c) $1/2$ (d) 2
9. $\sin 47^\circ + \sin 61^\circ - \sin 11^\circ - \sin 25^\circ$ is equal to
 (a) $\sin 36^\circ$ (b) $\cos 36^\circ$ (c) $\sin 7^\circ$ (d) $\cos 7^\circ$
10. If $\cos A = m \cos B$, then $\cot \frac{A+B}{2} \cot \frac{B-A}{2} =$
 (a) $\frac{m-1}{m+1}$ (b) $\frac{m+2}{m-2}$ (c) $\frac{m+1}{m-1}$ (d) none of these
11. If A, B, C are in A.P., then $\frac{\sin A - \sin C}{\cos C - \cos A} =$
 (a) $\tan B$ (b) $\cot B$ (c) $\tan 2B$ (d) none of these
12. If $\sin(B+C-A), \sin(C+A-B), \sin(A+B-C)$ are in A.P., then $\cot A, \cot B, \cot C$ are in
 (a) GP (b) HP (c) AP (d) none of these
13. If $\sin x + \sin y = \sqrt{3}(\cos y - \cos x)$, then $\sin 3x + \sin 3y =$
 (a) $2 \sin 3x$ (b) 0 (c) 1 (d) none of these
14. If $\tan \alpha = \frac{x}{x+1}$ and $\tan \beta = \frac{1}{2x+1}$, then $\alpha + \beta$ is equal to
 (a) $\pi/2$ (b) $\pi/3$ (c) $\pi/6$ (d) $\pi/4$

ANSWERS

1. (d) 2. (b) 3. (b) 4. (a) 5. (b) 6. (b) 7. (a) 8. (b)
 9. (d) 10. (c) 11. (b) 12. (b) 13. (b) 14. (d)

SUMMARY

- (i) $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$
 (ii) $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$
 (iii) $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$
 (iv) $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$
- (i) $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$
 (ii) $\sin C - \sin D = 2 \sin \frac{C-D}{2} \cos \frac{C+D}{2}$
 (iii) $\cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$
 (iv) $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$

CHAPTER 9

TRIGONOMETRIC RATIOS OF MULTIPLE AND SUBMULTIPLE ANGLES

9.1 INTRODUCTION

In this chapter, we intend to express the trigonometric ratios of multiple angles $2A, 3A, 4A, \dots$ etc. in terms of the trigonometric ratios of angle A and the trigonometric ratios of angle A in terms of the trigonometric ratios of sub-multiple angles $A/2, A/3, A/4 \dots$ etc. These results will be used to find the trigonometric ratios of some important angles viz. $18^\circ, 36^\circ, 54^\circ, 7\frac{1}{2}^\circ, 11\frac{1}{4}^\circ$ etc.

9.2 TRIGONOMETRIC RATIOS OF ANGLE $2A$ IN TERMS OF THAT OF ANGLE A

THEOREM 1 For the values of angle A for which the two sides are meaningful prove that:

- (i) $\sin 2A = 2 \sin A \cos A$ (ii) $\cos 2A = \cos^2 A - \sin^2 A$
(iii) $\cos 2A = 2 \cos^2 A - 1$ or, $1 + \cos 2A = 2 \cos^2 A$
(iv) $\cos 2A = 1 - 2 \sin^2 A$ or, $1 - \cos 2A = 2 \sin^2 A$
(v) $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$ (vi) $\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$
(vii) $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$

PROOF (i) We know that

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\Rightarrow \sin 2A = \sin A \cos A + \cos A \sin A$$

[Replacing B by A]

$$\Rightarrow \sin 2A = 2 \sin A \cos A$$

(ii) We know that

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\Rightarrow \cos 2A = \cos A \cos A - \sin A \sin A$$

[Replacing B by A]

$$\Rightarrow \cos 2A = \cos^2 A - \sin^2 A$$

(iii) We have,

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\Rightarrow \cos 2A = \cos^2 A - (1 - \cos^2 A)$$

$$\Rightarrow \cos 2A = 2 \cos^2 A - 1$$

$$\text{Again, } \cos 2A = 2 \cos^2 A - 1$$

$$\Rightarrow 1 + \cos 2A = 2 \cos^2 A$$

(iv) We have,

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\Rightarrow \cos 2A = (1 - \sin^2 A) - \sin^2 A$$

$$\Rightarrow \cos 2A = 1 - 2 \sin^2 A$$

$$\text{Again, } \cos 2A = 1 - 2 \sin^2 A$$

$$\Rightarrow 1 - \cos 2A = 2 \sin^2 A$$

(v) We know that

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\Rightarrow \tan 2A = \frac{\tan A + \tan A}{1 - \tan A \tan A}$$

[Replacing B by A]

$$\Rightarrow \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

(vi) We have,

$$\sin 2A = 2 \sin A \cos A$$

$$\Rightarrow \sin 2A = \frac{2 \sin A \cos A}{1}$$

$$\Rightarrow \sin 2A = \frac{2 \sin A \cos A}{\sin^2 A + \cos^2 A}$$

[$\because \sin^2 A + \cos^2 A = 1$]

$$\Rightarrow \sin 2A = \frac{2 \sin A \cos A}{\sin^2 A + \cos^2 A}$$

[Dividing Numerator and Denominator by $\cos^2 A$]

$$\Rightarrow \sin 2A = \frac{\cos^2 A}{\sin^2 A + \cos^2 A}$$

$$\Rightarrow \sin 2A = \frac{\cos A}{\frac{\sin^2 A}{\cos^2 A} + \frac{\cos^2 A}{\cos^2 A}}$$

$$\Rightarrow \sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$$

(vii) We have,

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\Rightarrow \cos 2A = \frac{\cos^2 A - \sin^2 A}{1}$$

$$\Rightarrow \cos 2A = \frac{\cos^2 A - \sin^2 A}{\cos^2 A + \sin^2 A}$$

[Replacing 1 by $\cos^2 A + \sin^2 A$]

$$\Rightarrow \cos 2A = \frac{\cos^2 A - \sin^2 A}{\cos^2 A + \sin^2 A}$$

[Dividing Numerator and Denominator by $\cos^2 A$]

$$\Rightarrow \cos 2A = \frac{\cos^2 A - \sin^2 A}{\cos^2 A + \sin^2 A}$$

$$\Rightarrow \cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$